Tutorial 7

t - Tests

An easy video tutorial on:

SPSS https://www.youtube.com/watch?v=ZpwZS3XnEZA

PSPP https://www.youtube.com/watch?v=NqS6DTXeXC4

Access to t distribution table http://www.ttable.org/

T Value (Critical Value) Calculator

Activity 1

Facebook wanted to find how many hours people spend using Facebook. They collected for a week the number of hours 100 males and 100 females spend on Facebook. This data sample set can be found on a spreadsheet names Tutorial7_DataSet.xlsx on Blackboard.

Using these data to attempt the following:

1. Form a research question, which examines the difference between the two groups.

   RQ: Is there a difference in the time male and female spend on Facebook?
2. Form a hypothesis, which examines the difference between the two groups.

   H_o: There is no significant difference in the time male and female spend on Facebook.

   H_a: There is significant difference in the time male and female spend on Facebook.

   H_o: $\bar{x}_{fb_time_male} - \bar{x}_{fb_time_female} = 0$ (two-tailed test)

   H_a: $\bar{x}_{fb_time_male} - \bar{x}_{fb_time_female} \neq 0$
3. What is the dependent variable?

   Time spend on facebook
4. What is the independent variable?

   The scores of male and female
5. What type is your data?

   Ratio - Scale
6. What is the experimental design?

   Between-group
7. If the data is normally distributed what statistical test could you use for significance?

   Independent sample t Test
8. Organise the data in SPSS or PSPP:

- In **Variable View** (where we set up the independent variable and dependent variable):
  - Create an appropriate number of variables
  - **In Data View** (where we set up the observations for each variable), insert the data from the Tutorial7_DataSet.xlsx spreadsheet (you will need to convert the excel spreadsheet to .csv format)
  - Save the .sav file, as 'FacebookStudy'.
  - Select Analyze > Descriptive Statistics > Explore, in Statistics set the Confidence Interval to 95%. In plots select Stem-and-leaf and histogram.
  - What are the results that you get from running Descriptive Statistics?

Descriptives

	Participants			Statistic	Std. Error
Hours Spend on Facebook	Male	Mean		**16.15**	.816
		95% Confidence Interval for Mean	Lower Bound	14.53	
			Upper Bound	17.77	
		5% Trimmed Mean		16.18	
		Median		15.00	
		Variance		66.634	
		Std. Deviation		8.163	
		Minimum		1	
		Maximum		30	
		Range		29	
		Interquartile Range		13	
		Skewness		.033	.241
		Kurtosis		-1.099	.478
	Female	Mean		**12.51**	.736
		95% Confidence Interval for Mean	Lower Bound	11.05	
			Upper Bound	13.97	
		5% Trimmed Mean		12.46	
		Median		12.00	
		Variance		54.131	
		Std. Deviation		7.357	
		Minimum		1	
		Maximum		25	
		Range		24	
		Interquartile Range		13	
		Skewness		.073	.241
		Kurtosis		-1.269	.478

Interpretation of Statistics

1. **Central Tendency (Mean, Median, Trimmed Mean)**:

- The **mean hours** spent on Facebook is higher for **males** (16.15 hours) than for **females** (12.51 hours), indicating that, on average, males spend more time on Facebook.
- The **median** hours are 15 for males and 12 for females, suggesting that the midpoint of hours spent is also higher for males.
- The **5% trimmed mean** (16.18 for males, 12.46 for females) is very close to the actual mean in both groups, suggesting minimal impact of outliers on the mean.

2. **Dispersion (Variance, Standard Deviation, Range, Interquartile Range)**:

- **Variance** and **standard deviation** are higher for males (variance = 66.634, SD = 8.163) compared to females (variance = 54.131, SD = 7.357), indicating greater variability in hours spent on Facebook among males.
- Both groups have the same **interquartile range** (IQR = 13), suggesting similar spread for the middle 50% of data.
- The **range** is slightly larger for males (29) compared to females (24), showing that the total spread of hours is wider among males.

3. **Skewness and Kurtosis**:

- As a general rule of thumb:
    1. If skewness is less than -1 or greater than 1, the distribution is highly skewed.
    2. If skewness is between -1 and -0.5 or between 0.5 and 1, the distribution is approximately symmetric or moderately skewed.
    3. If skewness is between -0.5 and 0.5, the distribution is approximately symmetric.
- **Skewness**: Both distributions have very low skewness (0.033 for males and 0.073 for females), indicating that the data for hours spent on Facebook is approximately symmetric for both genders.
- If the kurtosis is within the range of -0.5 and 0.5, the distribution is approximately symmetric.
- **Kurtosis**: Both groups show negative kurtosis (−1.099 for males and −1.269 for females), indicating a **platykurtic distribution** (flatter than a normal distribution) with lighter tails. This suggests fewer extreme values in both distributions.

4. **Confidence Interval for the Mean**:

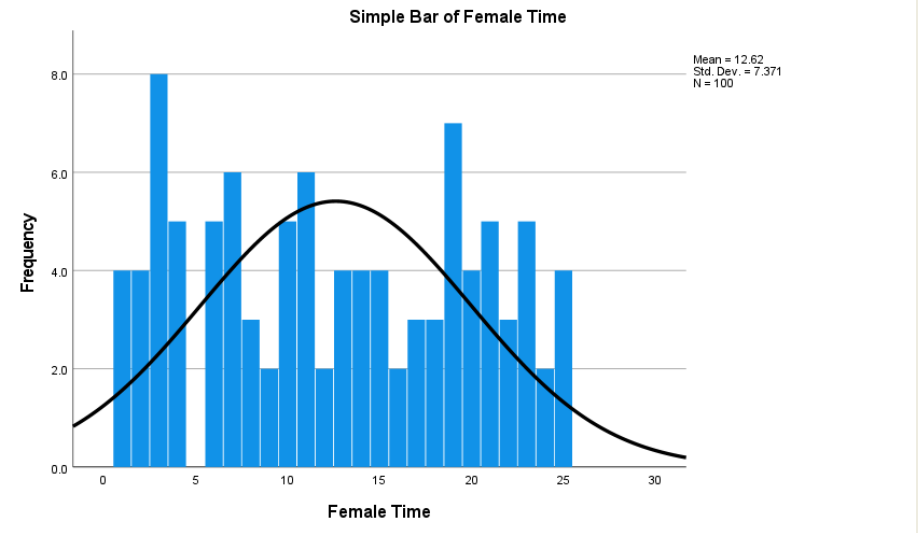
- The **95% confidence interval** for males is between 14.53 and 17.77 hours, and for females, it is between 11.05 and 13.97 hours. These intervals suggest that, with 95% confidence, the true mean hours spent on Facebook for males and females fall within these ranges.

Overall Insights

- **Higher Average Usage by Males**: Males spend more time on Facebook on average than females, with a higher mean and slightly greater variability.
- **Symmetry and Flatness**: Both groups have nearly symmetric distributions with a slightly flat (platykurtic) shape, indicating few extreme outliers.
- **Consistency in Spread**: While males have a larger range and standard deviation, the interquartile range (IQR) is consistent across both groups, indicating similar central data spread despite differences in overall variability.

To see this from the graphs. If you want to plot a graph with a normal curve create two separate variables for male time and female time and plot the graphs.





Graphs description

Histogram for Males

- **Shape**: The histogram for male participants would likely be approximately **symmetric**, as indicated by a very low skewness value (0.033). This means the bars would be fairly balanced on either side of the center.
- **Peak and Spread**:
    - The **mean** is 16.15, and the **median** is 15, so the peak of the histogram would likely be centered around these values.
    - With a **standard deviation of 8.163**, we can expect a moderately widespread around the mean, with bars extending to both the left and right.
- **Range and Extreme Values**:
    - The range is 29, with values spanning from a minimum of 1 hour to a maximum of 30 hours.
    - However, due to the negative kurtosis (-1.099), which indicates a **platykurtic distribution** (flatter than a normal distribution), the histogram would appear somewhat flatter and less peaked, with fewer extreme values or pronounced tails.
- **Interpretation**:
    - The histogram would show most male participants spending between approximately 10 to 20 hours on Facebook, with a fairly even distribution around the mean.
    - The flatness suggests that while most values cluster around the middle, there isn't a strong peak, and the distribution is spread out over a wider range.

Histogram for Females

- **Shape**: The histogram for female participants would also be **symmetric**, as indicated by the low skewness (0.073). This symmetry means that the bars on the left and right side of the center would be fairly balanced.
- **Peak and Spread**:
    - The **mean** is 12.51, and the **median** is 12, so the peak of the histogram would likely be centered around 12 hours.
    - The **standard deviation is 7.357**, slightly lower than for males, indicating a slightly narrower spread around the mean.
- **Range and Extreme Values**:

  - The range for females is 24, with hours spanning from a minimum of 1 hour to a maximum of 25 hours.
  - With a kurtosis of -1.269 (also indicating a **platykurtic distribution**), the histogram for females would be somewhat flat, similar to that of males, with fewer extreme values or outliers.

- **Interpretation**:
  - The histogram for females would show most participants spending between approximately 8 to 16 hours on Facebook, with a peak around the 12-hour mark.
  - Like the males, the flatness (platykurtic) means there would be a wider, less peaked distribution, reflecting less clustering around a single central value.

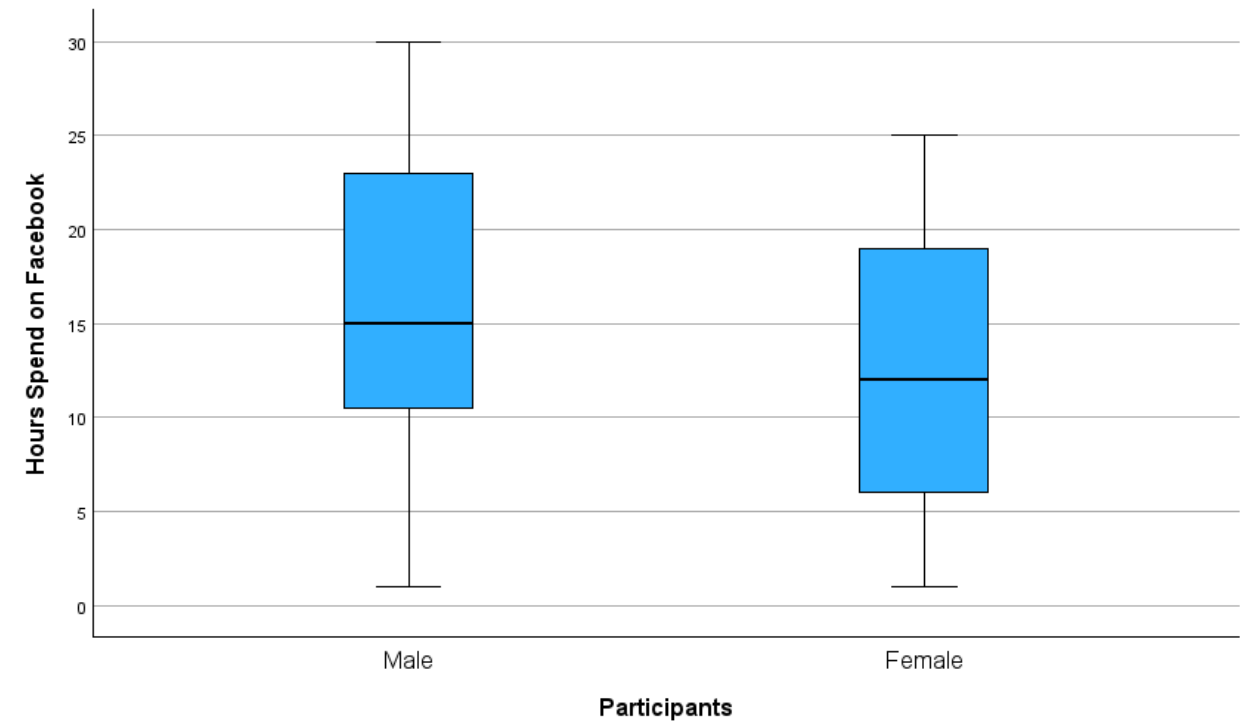
Comparison of the Two Histograms

- **Central Tendency**: Both histograms would have their peaks around the center of their respective distributions (around 15–16 hours for males and 12 hours for females), but the peak for males would be slightly shifted to the right due to the higher average time spent on Facebook.
- **Variability and Spread**: The male histogram would have a wider spread due to the higher standard deviation and range, showing a broader distribution of hours spent on Facebook compared to females.
- **Flatness (Platykurtic Shape)**: Both histograms would be relatively flat (platykurtic), with a lower peak and less pronounced tails, suggesting that the data is more evenly spread across a range without strong clustering around the mean.

In summary:

- The **male histogram** would show a symmetric, flat distribution with a peak around 15–16 hours, and a spread that extends over a wider range (1 to 30 hours).
- The **female histogram** would also be symmetric and flat, with a peak around 12 hours and a slightly narrower spread (1 to 25 hours).

Both distributions indicate a consistent use of Facebook hours with moderate variability and an even spread, without significant clustering or outliers.



Interpretation of the Stem-and-Leaf Plot for Males

- **Central Tendency**:
  - The **mean** for males is 16.15, and the **median** is 15.00. In a stem-and-leaf plot, this would be evident by a concentration of leaves around 15 and 16, with most values clustering in this area.
- **Spread and Variability**:
  - The **standard deviation** is 8.163, with a **range** of 29 (from 1 to 30). This suggests that the leaves are spread out across a broad range on the stem-and-leaf plot, extending from the low end (1) to the high end (30).
  - The plot would show a fairly wide spread of values, with a concentration near the middle but also some leaves extending towards the higher end.
- **Symmetry**:
  - With a **skewness** of 0.033, the distribution for males is nearly symmetric, so the stem-and-leaf plot would appear balanced around the central values (around 15–16). The leaves would not bunch up towards one side but would be distributed fairly evenly on both sides of the median.
- **Flatness (Platykurtic Shape)**:

- The **negative kurtosis** (-1.099) indicates a flat distribution, so the stem-and-leaf plot would likely show a relatively even spread of leaves rather than a sharp concentration in one area.

Interpretation of the Stem-and-Leaf Plot for Females

- **Central Tendency**:
  - The **mean** for females is 12.51, and the **median** is 12.00, so we would expect a concentration of leaves around the values 12 and 13 on the stem-and-leaf plot, with most values clustering around this central point.
- **Spread and Variability**:
  - With a **standard deviation** of 7.357 and a **range** of 24 (from 1 to 25), the leaves on the stem-and-leaf plot would cover a slightly narrower range than for males, extending from low values (1) up to the maximum of 25.
  - Although the range is smaller, there would still be a fair spread across the stems, with most leaves clustering near the center.
- **Symmetry**:
  - The **skewness** for females is 0.073, also indicating a nearly symmetric distribution. In the stem-and-leaf plot, the leaves would be fairly balanced around the median value of 12, showing no major shift towards higher or lower values.
- **Flatness (Platykurtic Shape)**:
  - The **kurtosis** of -1.269 suggests a flat distribution, so the leaves would be more evenly spread without a sharp peak. This would give the plot a somewhat "flat" look with a moderate spread of values around the central area.

Comparison of the Stem-and-Leaf Plots for Males and Females

- **Distribution**:
  - Both plots would show a concentration of values around the central tendencies, but males would have a higher concentration around 15–16, while females would concentrate around 12–13.
- **Range and Spread**:
  - The stem-and-leaf plot for males would have a wider spread, extending up to 30, while the plot for females would only go up to 25, reflecting the broader range of Facebook usage among males.
- **Symmetry**:

- Both plots would be nearly symmetric, with a balanced distribution of leaves around the center.

- **Flatness**:
  - Both plots would have a more evenly spread appearance, without a sharp concentration of leaves, due to the flat, platykurtic shape of both distributions.

Overall Summary

The stem-and-leaf plots for **Hours Spent on Facebook** would show a symmetric, evenly spread distribution for both males and females, with males generally spending more time on Facebook (higher central tendency) and having a wider range of usage. The plots would be nearly balanced around their respective medians, with moderate spreads and no sharp peaks, indicating a flat distribution.

- Answer the following questions:
  - Did we split the sample in this type of study according to treatment?

Yes

  - Is this type of study a 'between-subjects design' or a 'within-subjects design'? To answer the last question, perform research using 'Google' regarding the terms: 'between-subjects design' and 'within-subjects design'.

Between-subjects

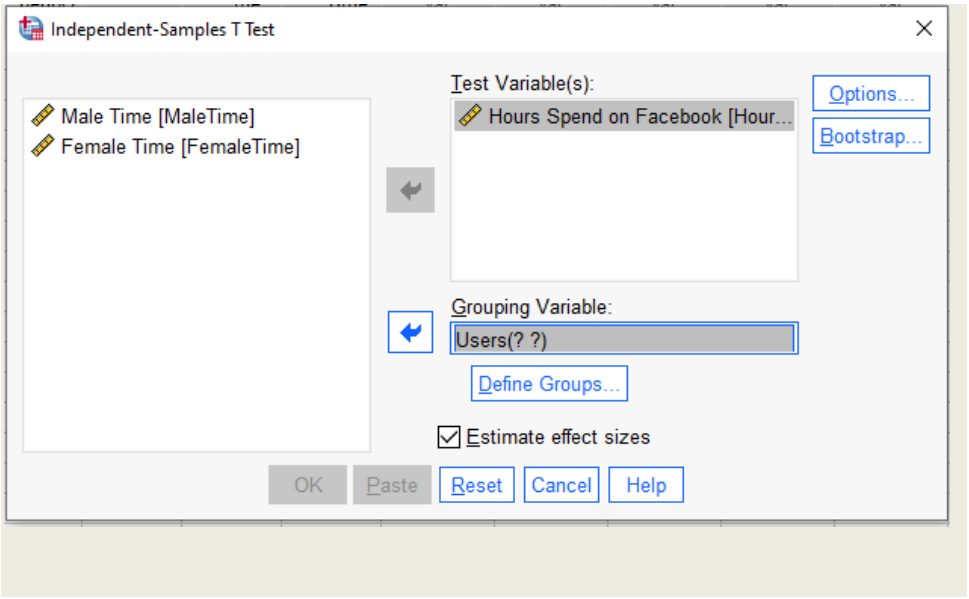
- Is the data normally distributed?

Based on skewness and the histogram with the normality curve yes

- What statistics test can you run to test the significance of the results?

On the basis that our date is radio scale, the study is between groups, the sample is large and the data is normally distributed we can run Independent sample t test

- Go to Analyse Compare Means and select the appropriate statistical test.


Independent-Samples T Test
Male Time [MaleTime]
Female Time [FemaleTime]
Test Variable(s):
Hours Spend on Facebook [Hour...
Options...
Bootstrap...
Grouping Variable:
Users(? ?)
Define Groups...
Estimate effect sizes
OK
Paste
Reset
Cancel
Help

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance		Mean	Std. Error	95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Difference	Difference	Lower	Upper
Hours Spend on Facebook	Equal variances assumed	1.391	.240	3.312	198	<.001	**.001**	3.640	1.099	**1.473**	**5.807**
	Equal variances not assumed			3.312	195.900	<.001	**.001**	3.640	1.099	1.473	5.807

- Interpret the results

We examine the p-value

A p-value is less than a 0.05 is considered significant

"There is less that 5% chance that the difference between the two means is due to chance"

From the table above (SPSS independent t Test results) we get that the two-tailed P value equals to **0.001**

By conventional criteria, the difference is considered to be very statistically significant.

Confidence Interval

The mean of Males minus the Females equals 3.64

95% confidence interval of difference: from 1.47 to 5.8

Intermedia values used in calculations

T = 3.3123

Df = 198

The returned t value is 3.3123, which is higher than the t value for the specific degree of freedom (df = 198) at the 95% confidence interval (t = 2.8388) (you can find this if you use the t score table

Standard error of difference = 1.099

(t = 3.312, p = **0.001)**

- Report the results
  - Short recap of method
  - Descriptive recap with standard deviations
  - Significant test (with t-statistics and p-value)
  - Hypothesis test recap

Two hundred **random** sample students were asked how many hours a week they spend on Facebook. An independent samples t-test revealed that Males (M = 16.15, SD = 8.16) spend significantly more time than Females (M = 12.51, SD = 7.35) on Facebook per week (t = 3.312, p = **0.001 h.s.).**

In both cases, the t-value is 3.312 with a p-value < .001. Since the p-value is less than 0.05, we reject the null hypothesis of no difference in mean hours spent on Facebook between the groups (or you could say the null hypothesis was rejected and the alternative hypothesis (or Hypothesis 1) was confirmed). This indicates a statistically significant difference between the groups.

Activity 2

The University introduced a new tool in Blackboard to enhance student revision. A lecturer used this tool experimentally with her students to test its effect on learning. She used 15 students as sample and tested their learning on the topics they covered in week 5 without making use of the new tool for revision. Then she asked her students to start using this tool and at the end of week 10 she tested her student's learning again. The table below contains the study results that can be used to determine whether the tool is effective. The maximum score that the students could achieve in both tests is 300.

Students	Scores Week 5	Scores Week 10
1	197	210
2	195	205
3	191	193
4	174	182
5	236	259
6	226	239
7	157	164
8	196	197
9	201	222
10	196	211
11	181	187
12	164	175
13	181	186
14	229	243
15	231	246

Using these data to attempt the following:

9. Form a research question, which examines the difference between the two groups.

   RQ: Is there a difference in the students learning before and after using the new tool for support revision?

10. Form a hypothesis, which examines the difference between the two groups.

    H_o: There is no significant difference in student test scores without and with the use of the revision tool in Blackboard.

    H_a: There is significant difference in student test scores without and with the use of the revision tool in Blackboard.

    H_o: $\bar{x}_{scores_without_using_revision_tool} - \bar{x}_{scores_after_using_revision_tool} = 0$ (two-tailed test)

    H_a: $\bar{x}_{scores_without_using_revision_tool} - \bar{x}_{scores_after_using_revision_tool} \neq 0$

11. What is the dependent variable?

    Students learning scores

12. What is the independent variable?

    The use of the tool or not

13. What type is your data?

    Ratio - Scale

14. What is the experimental design?

    Is this type of study a 'between-subjects design' or a 'within-subjects design'? Did we split the sample in this type of study according to treatment?

    Within-group, the same group has been exposed to treatment A and then treatment B

15. If the data is normally distributed what statistical test could you use for significance?

    Paired sample t Test

16. Organise the data in SPSS or PSPP:
    - In **Variable View** (where we set up the independent variable and dependent variable):
        - Create an appropriate number of variables
        - **In Data View** insert the data from the table above.
        - Save the .sav file, as 'Student_Scores'.
        - Select Analyze > Descriptive Statistics > Frequencies, for all measures of central tendency. In charts select a histogram with normal curve.
        - What are the results that you get from running Statistics?

Statistics

		Scores_Before	Scores_After
N	Valid	15	15
	Missing	0	0
Mean		**197.00**	**207.93**
Median		195.67[a]	205.00[a]
Mode		181[b]	164[b]
Std. Deviation		**24.393**	**28.562**
Variance		595.000	815.781
Skewness		**.201**	**.360**
Std. Error of Skewness		.580	.580
Range		79	95
Minimum		157	164
Maximum		236	259
Sum		2955	3119

a. Calculated from grouped data.
b. Multiple modes exist. The smallest value is shown

Interpretation of Statistics

1. **Central Tendency (Mean, Median, Mode)**:
    - The **mean score** increased from 197.00 (Scores Before) to 207.93 (Scores After), suggesting an improvement in scores after the intervention.

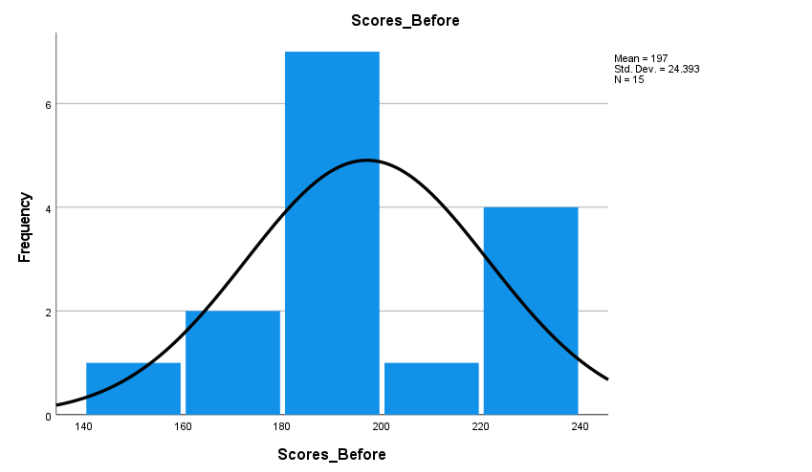
- The **median** values also show an increase, from 195.67 to 205.00, which is consistent with the mean and suggests that the central tendency of scores shifted upward after the intervention.
- **Modes**: The most frequent score was 181 before the intervention and 164 after, though the note indicates that multiple modes exist in both sets, and the smallest mode is shown.

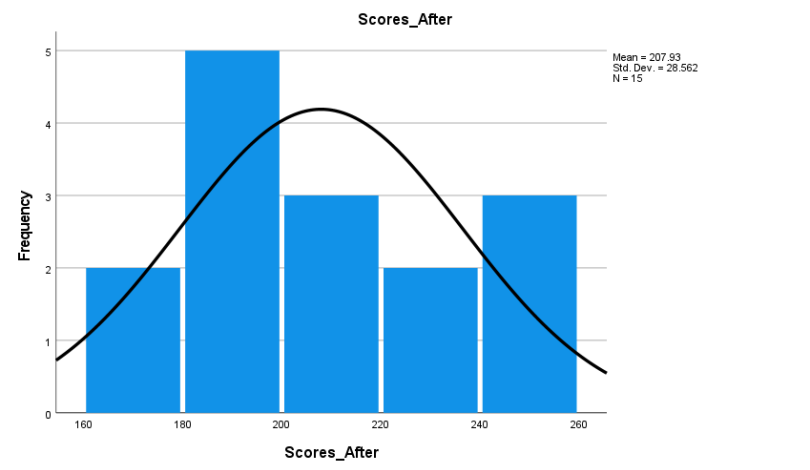
2. **Dispersion (Standard Deviation, Variance, Range)**:
   - **Standard Deviation**: The standard deviation increased from 24.393 (before) to 28.562 (after), indicating slightly more variability in scores after the intervention.
   - **Variance**: The variance also increased (from 595.000 to 815.781), further confirming an increase in score variability after the intervention.
   - **Range**: The range widened from 79 (before) to 95 (after), with the minimum score increasing from 157 to 164 and the maximum increasing from 236 to 259, suggesting that while overall scores improved, the spread between the lowest and highest scores also grew.
3. **Skewness**:
   - As a general rule of thumb:
     1. If skewness is less than -1 or greater than 1, the distribution is highly skewed.
     2. If skewness is between -1 and -0.5 or between 0.5 and 1, the distribution is approximately symmetric or moderately skewed.
     3. If skewness is between -0.5 and 0.5, the distribution is approximately symmetric.
   - **Skewness** values for both distributions are positive but low (0.201 for Scores Before and 0.360 for Scores After), indicating that both distributions are approximately symmetric, with a slight rightward skew.
   - This mild positive skew suggests that a few higher scores are pulling the distribution slightly to the right, but this effect is minimal.
4. **Sum of Scores**:
   - The total sum of scores increased from 2955 (before) to 3119 (after), reflecting the overall improvement in scores after the intervention.

Summary

- The **increase in mean and median scores** suggests an improvement in performance or results after the intervention.

- **Increased variability** (higher standard deviation, variance, and range) indicates that scores became more spread out after the intervention, with both higher maximum and minimum values.
- **Slight positive skewness** in both distributions suggests a nearly symmetric distribution with minimal skew.
- Overall, the data suggest a **positive effect** from the intervention, with scores generally improving and a broader distribution of scores after the intervention. This could indicate that while most participants improved, the degree of improvement varied.





Graphs description

Histogram for Scores Before

- **Shape**:
  - The **skewness** for Scores Before is 0.201, which indicates a **slight right skew**. However, this skewness is low, so the distribution would be close to symmetric, with a small tail extending to the right.

- **Peak and Central Tendency**:
    - The **mean** score is 197.00, and the **median** is 195.67, so the peak of the histogram would likely be centered around 195–200.
    - The presence of **multiple modes**, with the smallest mode at 181, suggests that the histogram might show one or more peaks near this value, but they would be close to the center.
- **Spread and Variability**:
    - The **standard deviation** is 24.393, and the **range** is 79 (minimum of 157 and maximum of 236). This suggests that the bars would extend from the lower end around 157 to the higher end near 236, with most scores clustered around the mean.
- **Interpretation**:
    - The histogram for Scores Before would likely show a relatively symmetric distribution centered around 195–200, with scores spanning from 157 to 236. The slight right skew might show a few values in the higher range, but the overall shape would be close to a bell curve.

Histogram for Scores After

- **Shape**:
    - The **skewness** for Scores After is 0.360, which is also a low positive skew, indicating a distribution that is **nearly symmetric** but with a small right tail.
- **Peak and Central Tendency**:
    - The **mean** score is 207.93, and the **median** is 205.00, so the peak of the histogram would be centered slightly to the right compared to Scores Before, around 205–210.
    - Like Scores Before, there are **multiple modes**, with the smallest mode at 164, but these values are near the center, suggesting a similar clustering of values around the mean.
- **Spread and Variability**:
    - The **standard deviation** of 28.562 and the **range** of 95 (minimum of 164 and maximum of 259) indicate greater variability in Scores After. This would be reflected in a slightly wider spread of bars, with the histogram extending from around 164 to 259.
- **Interpretation**:
    - The histogram for Scores After would show a slightly broader distribution, centered around 205–210, with more spread than Scores Before. The positive skew would create a slight tail on the right, indicating a few higher scores.

Comparison of the Two Histograms

- **Central Tendency**:
  - Both histograms would be centered near their respective means, with **Scores Before** peaking around 195–200 and **Scores After** peaking around 205–210, reflecting the overall improvement in scores.
- **Spread and Variability**:
  - The histogram for **Scores After** would show a wider spread, reflecting the increased standard deviation, variance, and range. This means that scores are more dispersed after the intervention, with some participants achieving much higher scores.
- **Skewness**:
  - Both histograms would be nearly symmetric, with a slight right skew. This is represented by a small rightward tail in both histograms, indicating that a few participants had scores significantly above the central values.

Overall Interpretation

- The histogram for **Scores Before** would be narrower, centered around 195–200, with minimal skew and a relatively symmetric distribution.
- The histogram for **Scores After** would be wider, centered around 205–210, with slightly more variability and a broader spread of scores, indicating that while scores improved on average, there was a greater range of improvement among participants.

In summary, the **Scores Before** histogram would be relatively symmetric and concentrated around the center, while the **Scores After** histogram would show a broader distribution with some higher scores, reflecting the increased mean and variability after the intervention.

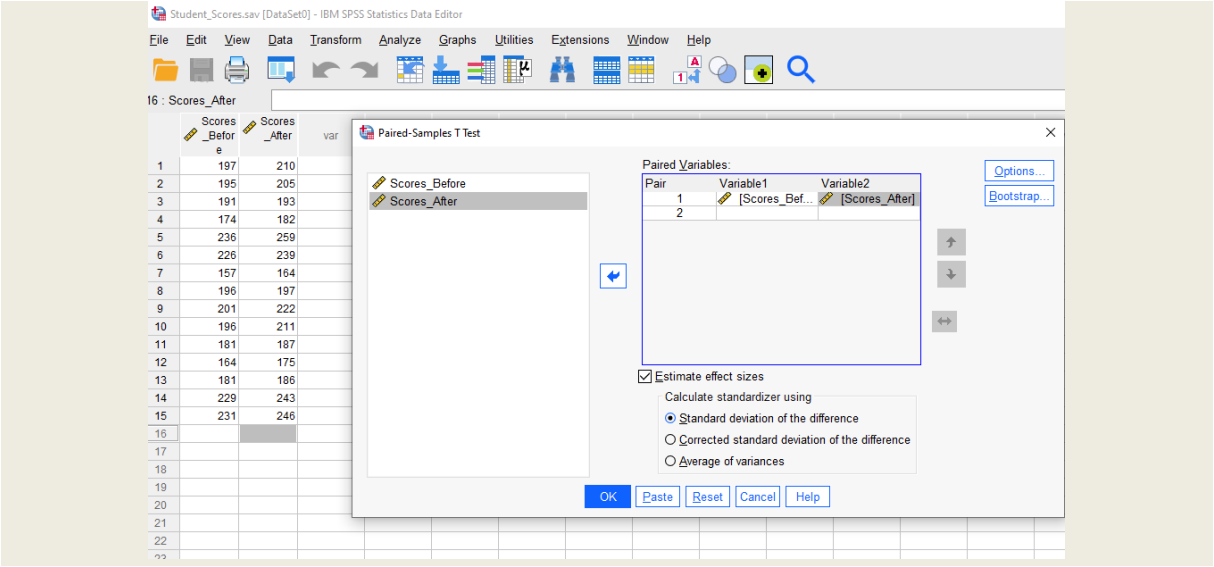
- Is the data normally distributed?

  Based on skewness and the histogram with the normality curve yes

- What statistics test can you run to test the significance of the results?

  On the basis that our date is radio scale, the study is within groups, the sample is normally distributed we can run paired sample t test

- Go to Analyse Compare Means and select the appropriate statistical test.


Student_Scores.sav [DataSet0] - IBM SPSS Statistics Data Editor
File Edit View Data Transform Analyze Graphs Utilities Extensions Window Help
16 : Scores_After
Scores_Before
Scores_After
var
1 197 210
2 195 205
3 191 193
4 174 182
5 236 259
6 226 239
7 157 164
8 196 197
9 201 222
10 196 211
11 181 187
12 164 175
13 181 186
14 229 243
15 231 246
Paired-Samples T Test
Scores_Before
Scores_After
Paired Variables:
Pair Variable1 Variable2
1 [Scores_Bef... [Scores_After]
2
Options...
Bootstrap...
Estimate effect sizes
Calculate standardizer using
Standard deviation of the difference
Corrected standard deviation of the difference
Average of variances
OK Paste Reset Cancel Help

T-Test

Paired Samples Statistics

		Mean	N	Std. Deviation	Std. Error Mean
Pair 1	Scores_Before	197.00	15	24.393	6.298
	Scores_After	207.93	15	28.562	7.375

- The **mean score** increased from **197.00 (Before)** to **207.93 (After)**, indicating an improvement in scores following the intervention.
- **Standard deviations** show moderate variability in both scores, with Scores After showing slightly more spread (28.562 vs. 24.393).
- The **standard error of the mean** is 6.298 for Scores Before and 7.375 for Scores After, indicating the precision of these mean estimates.

Paired Samples Correlations

		N	Correlation	Significance One-Sided p	Significance Two-Sided p
Pair 1	Scores_Before & Scores_After	15	.984	<.001	<.001

- The correlation between Scores Before and Scores After is 0.984, with a p-value < .001. This extremely high positive correlation suggests a very strong relationship between the scores before and after the intervention. It indicates that participants' scores before the intervention are highly predictive of their scores afterward, suggesting consistency in performance trends across participants.

Paired Samples Test

		Paired Differences					t	df
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference Lower	Upper		
Pair 1	Scores_Before - Scores_After	**-10.933**	**6.330**	1.634	-14.439	-7.428	**-6.690**	**14**

Interpret the results

- We examine the p-value
- A p-value is less than a 0.05 is considered significant
- "There is less that 5% chance that the difference between the two means is due to chance"

- From the table above (SPSS independent t Test results) we get that the two-tailed P value equals to **0.001**
- By conventional criteria, the difference is considered to be very statistically significant.
- **Confidence Interval**
- The mean before using the tool for revision equals 197
- 95% confidence interval of difference: from -14.439 -7.428
- **Intermedia values used in calculations**
- **T = -6.690**
- **Df = 14**
- **Standard error of difference = 1.634**
- **(**t = **-6.690**, p = <**0.01)**
- **The returned t value is -6.690 (the negative simply means that it lies to the left of the mean), which is higher than the t value for the specific degree of freedom (df = 15) at the 95% confidence interval (t = 2.131) (you can find this if you use the t score table**

- The **mean difference** between Scores Before and Scores After is **-10.933**, meaning that, on average, scores increased by 10.933 points after the intervention.
- The **95% confidence interval** for the mean difference is from **-14.439 to -7.428**, indicating we are 95% confident that the true mean difference falls within this range. Since this interval does not include zero, it supports the conclusion that the difference is statistically significant.
- The **t-value** is **-6.690** with **14 degrees of freedom**, and the **p-value is < .001** (both one-sided and two-sided). This very low p-value indicates a highly significant difference between Scores Before and Scores After, allowing us to reject the null hypothesis that there is no difference.
    - Report the results
        - Short recap of method
        - Descriptive recap with standard deviations
        - Significant test (with t-statistics and p-value)
        - Hypothesis test recap

Fifteen students' **random** sample were given a test before (in week 5) and after (week 10) making use of a Blackboard tool to support revision. A paired samples t-test revealed that After Scores (M = 207.93, SD = 28.562) increased significantly (M = 197.00, SD = 24.393) after 5 weeks of making use of the revision tool (t = -6.690, p = <0.01 h.s.). T **Therefore, the null hypothesis was rejected, and the alternative hypothesis (or Hypothesis 1) was confirmed.**

Summary

- **Statistically Significant Increase**: There is a statistically significant increase in scores after the intervention, as indicated by the p-value < .001 and the confidence interval that excludes zero.
- **Strong Correlation**: The high correlation (0.984) between the two sets of scores suggests that participants who scored high or low before the intervention tended to do so afterward, though on average their scores improved.
- **Conclusion**: The intervention appears to have had a positive effect on scores, with a consistent and statistically significant improvement across participants. The results provide strong evidence that the mean score after the intervention is significantly higher than before.